

a Three-concentration reconstruction

194 chemicals · folds 1–4 · paired identity-group 95% intervals · AB = AnchorBoost

Our reference models

AnchorBoost stack

CPU single model

AnchorBoost alone

Tabular foundation models

Limix-2 + AB + CheMeleon

TabPFN-3.5 + AB + CheMeleon

TabPFN-3.5 + AB features

TabPFN-2 + AB features

TabICLv2

Neural processes and latent models

LPM, 20 seeds

CNP, released checkpoints (n=193)

TNP

LPM, 3 seeds

FlowNP, 3 seeds

TE-TNP

Molecular and structural models

KANO embedding + AB

MotIL embedding + AB

KANO, structure + three points

MotIL, structure + three points

ARCANet, structure only

KANO, structure only

MotIL, structure only

Classical curves and curve priors

DR-PFN, fine-tuned

DR-PFN, synthetic prior

Empirical GP

Gaussian process (n=193)

Hill empirical Bayes (n=193)

Low-rank and tensor models

TabImpute

SoftImpute

CP tensor

Toxicological read-across

GenRA, structure only

Interpolation and neighbours

Log-linear interpolation

Analog chemicals, nearest neighbors

SNN

N², DRNN

N², AutoNN



● Our win □ Tie / CI crosses zero ▼ Our loss — Saved at k = 3 only

Retrained CNP runs on its own training-only scale (k = 3 MAE 1.2363, 193 chemicals), so it is listed here, off the common axis.

b Our paired result

Columns: k = 1, 2, 3, 4 measured concentrations

